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Suspension of a triple-flow aircraft turbine engine

Abstract

A triple-flow turbine engine for an aircraft, this turbine engine having a longitudinal axis and including upstream suspension elements which are located in a first plane perpendicular to the axis and are connected or fixed to the gas generator of the turbine engine; downstream suspension elements which are located in a second plane perpendicular to the axis and are connected or fixed to the gas generator; and thrust-absorbing rods which include first ends which are connected or fixed to the gas generator and opposing second ends which are located in a third plane perpendicular to the axis, wherein the first, second and third planes are located at a cold compartment of the gas generator.

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Background/Summary

TECHNICAL FIELD OF THE INVENTION

(1) The present invention relates to the field of the suspension of a triple-flow turbine engine, in particular of aircraft.

TECHNICAL BACKGROUND

(2) The prior art comprises in particular the documents WO-A1-2010/092263, FR-A1-2 981 989, FR-A1-2 987 416, FR-A1-3 008 462, FR-A1-3 008 463, FR-A1-3 041 054, FR-A1-3 078 998 and U.S. Pat. No. 9,091,207 B2.

(3) As shown in FIG. 1, an aircraft turbine engine **10** generally comprises a gas generator **12** comprising at least one compressor **14**, an annular combustion chamber **16** and at least one turbine **18**.

(4) The turbine engine **10** further comprises at least one propulsion propeller **20** which is driven by a shaft of the turbine **18**.

(5) The gas generator **12** comprises an annular splitter nose **22** for separating two annular ducts V1, V2 of an internal primary flow F1 inside the gas generator **12**, and of an external secondary flow F2 around the gas generator **12** respectively.

(6) When the propulsion propeller **20** is located upstream of the gas generator **12** as in the example shown, it is called fan and its purpose is to accelerate the air drawn in. A part of the airflow exiting the fan flows around the gas generator **12** to form the secondary flow F2 which allows to generate most of the thrust produced by the turbine engine **10**. The remaining of the air flow exiting the fan enters the gas generator **12** to form the primary flow F1. This air is compressed in the or each compressor **14**, then mixed with fuel and burned in the combustion chamber **16**. The combustion gases are then expanded in the or each turbine **18** in order to rotate the rotor of the turbine and thus the propulsion propeller **20**.

(7) The propeller **20** comprises vanes **24** that generally have an angular position, referred to as pitch, that is stationary about an axis Y perpendicular to the longitudinal axis X of the turbine engine **10**. Even in the low speed phases of the engine, such as idling, this pitch is configured so that the flow rate of the secondary flow F2 generated is sufficient to feed fluid/air surface exchangers **26**, which may or may not be flat, installed in the duct for the flow of the secondary

flow F2. These exchangers **26** are needed to cool equipment of the turbine engine.

(8) FIGS. 2 and 3 show two possible configurations for the suspension of the turbine engine **10** of FIG. 1. A pylon **28** is used to attach and suspend the turbine engine **10** from a part of the aircraft, such as a wing **46** for example. The pylon **28** has a generally elongated shape along the longitudinal axis X of the turbine engine **10**.

(9) The turbine engine **10** comprises upstream suspension members **48** and downstream suspension members **50**. In the suspension configuration of FIG. 2, the downstream suspension members **50** are attached to a turbine casing **52** of the turbine engine **10**, and the upstream suspension members **48** are attached to an intermediate casing **54** of the turbine engine **10**. Thrust-absorbing rods **56** extend from the upstream suspension members **48** to the gas generator **12**.

(10) In the suspension configuration of FIG. 3, the downstream suspension members **50** are attached to the turbine casing **52** of the turbine engine **10**, and the upstream suspension members **48** are attached to a fan casing **58** of the turbine engine **10**. Thrust-absorbing rods **56** extend from the downstream suspension members **50** to the gas generator **12**.

(11) To improve the performance of the engine (higher thrust and lower fuel consumption), some engine architectures aim to increase the bypass ratio (BPR) by increasing the diameter of the propeller **20**. In order to limit the mass of the turbine engine **10**, the shroud **28** (with the fan casing **58**) located around the propulsion propeller **20** may be removed, and the vanes **24** of the propeller **20** may be made steerable, and thus variable in pitch, to control the thrust level of the turbine engine.

(12) However, it happens that in certain phases such as ground idle, the run-up or thrust reversal phases, the flow rate of the secondary flow F2 is no longer sufficient for the thermal management of the turbine engine which needs heat exchange between fluids and air in order to ensure the cooling of its equipment and its optimal operation.

(13) To address this problem, it has already been proposed to provide an additional bypass flow dedicated to the thermal management of the turbine engine.

(14) As shown in FIG. 4, the turbine engine **10** then comprises a secondary propeller **30** driven in rotation by a shaft of the gas generator **12**. This propeller **30** is located in the duct V1 for the flow of the primary flow F1 and upstream of a second annular splitter nose **32** separating two annular ducts V11, V12 for the respective flow of a first internal flow F11 and a second external flow F12.

(15) The exchangers **26** are housed in the flow duct V12 and are fed by the second external flow F12. The first internal flow F11 feeds the compressor **14**, as mentioned above.

(16) FIG. 4 allows to show that the gas generator **12** comprises two annular compartments C1, C2 which extend around the axis X.

(17) The first annular compartment C1 is called hot compartment because it is relatively close to the duct V11 and therefore more exposed to the heat generated by the compression, by the combustion chamber **16** and by the combustion gases. This hot compartment C1 extends axially between the second splitter nose **32** and a first nozzle **34** for ejecting the first internal flow F11 from the turbine or the turbines. This hot compartment C1 is further delimited radially by first and second annular walls **36**, **38**, respectively internal and external, which extend coaxially around each other. The first wall **36** externally defines the duct V11 for the flow of the first internal flow F11, and the second wall **38** internally defines the duct V12 for the flow of the second external flow F12.

(18) The second annular compartment C2 is called cold compartment because it is less exposed to heat, in particular because it is separated from the hot compartment by the duct V12. This cold compartment C2 extends axially between the first splitter nose **22** and a second nozzle **40** for ejecting the second external flow F12. This cold compartment C2 is further delimited radially by third and fourth annular walls **42**, **44**, respectively internal and external, which extend coaxially around each other. The third wall **42** externally defines the duct V12 for the flow of the second external flow F12, and the fourth wall **44** internally defines the duct V2 for the flow of the

secondary flow F2.

(19) Lot of equipment in the turbine engine **10** are installed in the hot compartment C1, which is large. The cold compartment C2 is relatively small and is reserved for equipment that cannot withstand the temperatures of the hot compartment C1, in particular the electronic equipment.

(20) This type of installation has several consequences: The equipment in the hot compartment C1 shall be thermally protected. These thermal protections constitute additional volumes and masses to be installed in an already very constrained environment, which tends to hinder the accessibility for the inspections and the maintenance of the turbine engine. A localized ventilation must be added in the hot compartment C1 for certain (electronic) equipment that is more sensitive to heat and cannot be installed in the cold compartment C2 because space is too limited there. The equipment has a reduced service lifetime in the hot compartment C1 and this restrictive environment has consequences for its design: the choice of their technologies is limited and its mass is increased.

(21) FIG. 5 shows a possible configuration for the suspension of the turbine engine **10** of FIG. 4. The upstream **48** and downstream **50** suspension members are located at the level of the cold compartment C2. The downstream suspension members **50** are connected to the intermediate casing **54** and the thrust-absorbing rods **56** extend from the gas generator **12** to an attachment point on the pylon **28** which is located well downstream of the suspension members **48**, **50**.

(22) FIG. 6 illustrates another type of triple-flow turbine engine **10**, in which the main propeller **20** is located downstream of the gas generator **12**. This main propeller **20** is part of a counter-rotating, non-shrouded propeller doublet.

(23) This FIG. 6 shows a possible configuration for the suspension of the turbine engine **10**. The upstream suspension members **48** are located at the level of the cold compartment C2, and the downstream suspension members **50** are located at the level of the hot compartment C1. The thrust-absorbing rods **56** extend from the gas generator **12** to an attachment point on the pylon **28** which is located between the suspension members **48**, **50**.

(24) The different suspension configurations mentioned above have constraints: the upstream and downstream suspension members **48**, **50** are located in planes P1, P2 perpendicular to the longitudinal axis X of the turbine engine **10**, which are located at a relatively small axial distance (called the center distance) from each other, which generates significant forces in the **20** members **48**, **50**; the distribution of the forces between the upstream and downstream suspension members **48**, **50** is also problematic because the center of gravity of the turbine engine **10** is not in the middle between the two suspension planes P1, P2; and **25** the thrust of the turbine engine **10**, which is generated in particular by the propeller **20**, passes through casings to be transmitted to the pylon **28** by the connecting rods **56**.

(25) The invention provides a solution allowing to solve some or all of these problems.

SUMMARY OF THE INVENTION

(26) The invention relates to a triple-flow turbine engine for an aircraft, this turbine engine having a longitudinal axis and comprising: a gas generator comprising at least one compressor, an annular combustion chamber, and at least one turbine, at least one main propeller rotated by a shaft of the gas generator, the gas generator comprising: a first annular splitter nose for separating two annular ducts for the respective flow of an internal primary flow inside the gas generator and of an external secondary flow around the gas generator, a second annular splitter nose for separating two annular ducts for the respective flow of a first internal flow and of a second external flow, a first nozzle for ejecting the first internal flow, a second nozzle for ejecting the second external flow, a first annular compartment, referred to as hot compartment, extending axially between the second splitter nose and the first nozzle, and between first and second annular walls, respectively internal and external, which extend, preferably coaxially, around each other, the first wall externally defining said duct for the flow of the first internal flow, and said second wall internally defining said duct for the flow of the second external flow, a second annular compartment, referred to as cold compartment, extending axially between the first splitter nose and the second nozzle, and between third and

fourth annular walls, respectively internal and external, which extend, preferably coaxially, around each other, the third wall externally defining said duct for the flow of the second external flow, and said fourth wall internally defining said duct for the flow of the secondary flow, upstream suspension members of the turbine engine, which are located in a first plane perpendicular to said axis and which are connected or attached to the gas generator, downstream suspension members of the turbine engine, which are located in a second plane perpendicular to said axis and which are connected or attached to the gas generator, and thrust-absorbing rods which comprise first ends connected or attached to the gas generator and opposite second ends located in a third plane perpendicular to said axis, characterized in that said first, second and third planes are located axially between the first splitter nose and the second nozzle, and in that at least one part of said upstream and downstream suspension members are connected or attached to said third wall.

(27) In the present application, the terms “upstream” and “downstream” refer to the flow direction of the gases in the turbine engine.

(28) The invention thus proposes to bring the suspension planes of the turbine engine axially closer to each other and to position them at the level of the third wall and thus of the cold compartment of the gas generator. The suspension members are connected or attached to the third wall, which is thus configured to transmit the forces associated with the weight and the thrust of the turbine engine to the pylon. This third wall is therefore at least partly structural, which means that it has a rigidity and a mechanical strength that allow it to ensure the transmission of the above-mentioned forces.

(29) Advantageously, the turbine engine is devoid of additional suspension members downstream of said downstream suspension members or of the thrust-absorbing rods so that the turbine engine is attached in a cantilevered manner to the pylon.

(30) The turbine engine according to the invention may comprise one or more of the following characteristics, taken in isolation from each other, or in combination with each other: the downstream suspension members are attached or connected to said third wall; the downstream suspension members are located in the cold compartment; the upstream suspension members are attached to the third wall and/or the fourth wall; the upstream suspension members are located just downstream of said first splitter nose; said thrust-absorbing rods extend from the upstream suspension members or downstream of these members to connection points on the gas generator, these connection points being located inside the third wall; said connection points are located upstream of said second splitter nose; alternatively, they could be located downstream of said second splitter nose; at least one part of said second and third walls are connected together by tubular reinforcements or arms, to form an assembly; the second and third walls comprise openings and removable hatches for closing these openings; said assembly forms a segment in one piece; said assembly comprises a first segment in one piece and a second segment which can be detached from the first segment and which can itself be in one piece or formed of two half-shells articulated with respect to each other about an axis substantially parallel to said longitudinal axis; at least one heat exchanger is located in said second external duct, and/or at least one equipment, such as an accessory box or gear box, is located in said cold compartment; said at least one main propeller is located upstream of the gas generator; said at least one main propeller is located downstream of the gas generator; the turbine engine comprises a secondary propeller driven in rotation by a shaft of the gas generator, this propeller being located in the duct for the flow of the primary flow and upstream of said second annular splitter nose; the pylon comprises an upstream end located between two adjacent stator vanes extending into the duct for the flow of the secondary flow, or located at the level of a root of one of these vanes; the upstream suspension members are located between two adjacent stator vanes extending in the secondary flow duct, or located at the level of a root of one of these vanes; the upstream suspension members are located downstream of the first splitter nose, and upstream or in line with the second splitter nose; said assembly is attached downstream of a casing located between two compressors of the gas generator.

(31) The present invention also relates to an assembly comprising a turbine engine as described above, and a pylon for hanging this turbine engine to an aircraft, the pylon having a generally elongated shape and being connected or attached respectively to the upstream and downstream suspension members of the turbine engine.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) Further characteristics and advantages will be apparent from the following description of a non-limiting embodiment of the invention with reference to the appended drawings in which:

(2) FIG. 1 is a schematic axial sectional view of a double-flow turbine engine for an aircraft,

(3) FIG. 2 is a schematic axial sectional view of the turbine engine of FIG. 1 and its suspension pylon to an aircraft, and illustrates a configuration of the suspension members,

(4) FIG. 3 is a schematic axial sectional view of the turbine engine of FIG. 1 and its suspension pylon to an aircraft, and illustrates an alternative configuration of the suspension members,

(5) FIG. 4 is a schematic half-view in axial section of a triple-flow turbine engine for an aircraft,

(6) FIG. 5 is a schematic axial sectional view of the turbine engine of FIG. 4 and its suspension pylon to an aircraft, and illustrates a configuration of the suspension members,

(7) FIG. 6 is a schematic axial sectional view of a downstream propulsion propeller turbine engine and its suspension pylon to an aircraft, and illustrates a configuration of the suspension members,

(8) FIG. 7 is a view similar to that of FIG. 5 and illustrates an embodiment of a turbine engine according to the invention and its suspension pylon to an aircraft,

(9) FIG. 8 is a schematic perspective view of an assembly for the turbine engine of FIG. 7,

(10) FIG. 9 is a schematic perspective view of an alternative embodiment of the assembly for the turbine engine of FIG. 7,

(11) FIG. 10 is a schematic view of an alternative embodiment of the assembly of FIG. 9, seen from the front from the downstream face,

(12) FIG. 11 is a view similar to that of FIG. 7 and illustrates an alternative embodiment of a turbine engine according to the invention and its suspension pylon to an aircraft,

(13) FIG. 12 is a view similar to that of FIG. 7 and illustrates another alternative embodiment of a turbine engine according to the invention and its suspension pylon to an aircraft,

(14) FIG. 13 is a view similar to that of FIG. 7 and illustrates another alternative embodiment of a turbine engine according to the invention and its suspension pylon to an aircraft.

(15) FIG. 14 is a view similar to that of FIG. 7 and illustrates another alternative embodiment of a turbine engine according to the invention and its suspension pylon to an aircraft,

(16) FIG. 15 is a view similar to that of FIG. 7 and illustrates another alternative embodiment of a turbine engine according to the invention and of its suspension pylon to an aircraft, and

(17) FIG. 16 is a similar view to FIG. 7 and illustrates another alternative embodiment of a turbine engine according to the invention and its suspension pylon to an aircraft.

DETAILED DESCRIPTION OF THE INVENTION

(18) FIGS. 1 to 6 have been described above.

(19) FIG. 7 represents an aircraft turbine engine 10 according to an embodiment of the invention, as well as a pylon 28 for suspending the turbine engine 10 to a part of an aircraft, such as a wing for example.

(20) The pylon 28 has a generally elongated shape along an axis Y that is parallel to the longitudinal axis X of the turbine engine 10. The pylon 28 comprises an external diameter noted Dext1 and an internal diameter noted Dint1.

(21) The turbine engine 10 is similar to that shown in FIG. 4. It comprises a gas generator 12 comprising at least one compressor 14, an annular combustion chamber 16 and at least one turbine

18.

(22) The gas generator **12** preferably comprises two successive compressors **14a**, **14b**. The compressor **14a** located upstream is a low pressure compressor. The compressor **14b** located downstream is a high pressure compressor. The two compressors **14a**, **14b** are separated from each other by an intermediate casing **54** or inter-compressor which comprises an internal hub **54a** and a ferrule **54b** extending around the hub **54a** and connected to the hub by arms not shown.

(23) The turbine engine **10** further comprises a propulsion propeller **20** which is located upstream of the gas generator **12** and is driven by a shaft of the turbine **18**. The propeller **20** has an external diameter noted Dext2 and an internal diameter noted Dint2.

(24) The gas generator **12** comprises an annular splitter nose **22** for separating two annular ducts **V1**, **V2** for the respective flow of an internal primary flow **F1** inside the gas generator **12**, and of an external secondary flow **F2** around the gas generator **12**.

(25) The turbine engine **10** comprises a secondary propeller **30** driven in rotation by a shaft of the gas generator **10**. This propeller **30** is located in the duct **V1** for the flow of the primary flow **F1** and upstream of a second annular splitter nose **32** separating two annular ducts **V11**, **V12** for the respective flow of a first internal flow **F11** and a second external flow **F12**.

(26) Exchangers **26** are located in the duct **V12** and are fed by the second external flow **F12**. The first internal flow **F11** feeds the compressor **14**, as mentioned above.

(27) The gas generator **12** comprises two annular compartments **C1**, **C2** which extend around the axis **X**.

(28) The first annular compartment **C1**, or hot compartment, extends axially between the second splitter nose **32** and a first nozzle **34** for ejecting the first internal flow **F11** from the turbine or the turbines. This hot compartment **C1** is further delimited radially by first and second annular walls **36**, **38**, respectively internal and external, which extend coaxially around each other. The first wall **36** externally defines the duct **V11** for the flow of the first internal flow **F11**, and the second wall **38** internally defines the duct **V12** for the flow of the second external flow **F12**.

(29) The second annular compartment **C2**, called cold compartment, extends axially between the first splitter nose **22** and a second nozzle **40** for ejecting the second external flow **F12**. This cold compartment **C2** is further delimited radially by third and fourth annular walls **42**, **44**, respectively internal and external, which extend coaxially around each other. The third wall **42** externally defines the duct **V12** for the flow of the second external flow **F12**, and the fourth wall **44** internally defines the duct **V2** for the flow of the secondary flow **F2**.

(30) The turbine engine **10** comprises, downstream of the propeller **20** and the splitter nose **22**, stator vanes **21** which extend radially outwards from the wall **44** and through the duct **V2**.

(31) The pylon **28** comprises a free end **28a** upstream. This free end **28a** is located here between two adjacent stator vanes **21** or at the level of the root of one of these vanes **21**. The free end **28a** of the pylon **28** is located between the leading and trailing edges of the stator vanes **21**.

(32) The cold compartment **C2** is designed to house equipment of the turbine engine **10**, as will be described in more detail below.

(33) The turbine engine **10** is suspended from the pylon **28** by means of upstream suspension members **48** and downstream suspension members **50**.

(34) The upstream suspension members **48** are located in a first plane **P1** perpendicular to the axis **X** and which are connected or attached to the gas generator **12**. The downstream suspension members **50** are located in a second plane **P2** perpendicular to the axis and are connected or attached to the gas generator **12**.

(35) According to the invention, these planes **P1**, **P2** are located axially between the first splitter nose **22** and the second nozzle **40**, i.e. at the level of the cold compartment **C2**. Furthermore, at least one part of these suspension members **48**, **50** are connected or attached to the wall **42**.

(36) In the example shown, Dext1 is less than Dext 2, and Dint1 is less than the maximum diameter of the wall **44** so that a lower part of the pylon **28** is housed in the cold compartment **C2**, as shown

in the figure.

(37) The one or more members **48** are located at the upstream end **28a** of the pylon **28** and may be located just downstream of the splitter nose **22**. They may be attached or connected to the wall **42** and/or to the wall **44**.

(38) The members **48** may be located substantially in line with the splitter nose **32**. In other words, the plane **P1** may pass substantially at the level of the splitter nose **32** or just upstream of the latter, as shown in the drawing.

(39) Thrust-absorbing rods **56** extend from the members **48** to the gas generator **12**. The connecting rods **56** extend radially outwardly from upstream to downstream and comprise first radially internal ends **56a** connected, for example by clevises, to the wall **42** or to points **W** of the gas generator **12** located radially inwardly of this wall **42**. These points **W** may be located upstream of the splitter nose **32**. However, this is not always the case, as illustrated by the alternatives **14** to **16** described below.

(40) The connecting rods **56** comprise second radially external ends **56b** which are located in a third plane **P3** perpendicular to the axis **X**. The plane **P3** is located between the planes **P1** and **P2**. These second ends **56b** are, for example, connected by a spreader to the pylon **28**, just downstream of the members **48**.

(41) The members **50** are located downstream of the members **48** and are attached or connected to the wall **42**. They are located inside the cold compartment **C2**.

(42) The wall **42** is connected to the wall **38** by tubular arms **60** which extend radially into the duct **V12** and are used for the passage of auxiliaries through this duct. The auxiliaries are, for example, electrical harnesses and cables, oil pipelines, ventilation conduits, etc. The walls **38**, **42** and the arms **60** form a whole called the assembly **62**.

(43) Alternatively, the arms **60** could be replaced by (non-tubular) reinforcements.

(44) The assembly **62** is intended to be attached to the downstream end of the casing **54**. The wall **42** has its upstream end attached to the downstream end of the ferrule **56b**, and the wall **38** has its upstream end attached to the hub **56a**.

(45) FIGS. **8** to **10** illustrate example embodiments of this assembly **62**.

(46) In the case of FIG. **8**, the assembly **62** is in one piece. The wall **42** comprises orifices **42a** which open into the internal cavities of the arms **60** for the passage of the aforementioned auxiliaries. The wall **42** further comprises openings **42b** that are configured to allow maintenance and inspection operations to be performed. These openings **42b** are intended to be sealed by removable hatches which are not shown. These hatches can be dismounted and removed through similar openings provided on the wall **44** for example.

(47) Similarly, the wall **38** may comprise similar openings **38a** to allow maintenance and inspection operations to the interior of the assembly **62**. The openings **42b**, **38a** in the walls **38**, **42** are preferably radially aligned with each other.

(48) The assembly **62** of FIG. **8** is structural and configured to transmit the forces from the turbine engine **10** to the pylon **28** during operation.

(49) In the case of FIG. **9**, the assembly **62** comprises two segments, respectively upstream **62a** and downstream **62b**. The upstream segment **62a** is similar to the assembly **62** in FIG. **8**. The downstream segment **62b** is intended to be detachably attached to the downstream end of the upstream segment **62a**. More specifically, the downstream segment **62b** comprises a portion of the wall **42** that is intended to be detachably attached to the portion of the wall **42** of the segment **62a**, and a portion of the wall **38** that is intended to be detachably attached to the portion of the wall **38** of the segment **62a**.

(50) The upstream **62a** and the downstream **62b** segments are each of the O-Duct type, i.e., they each form an uninterrupted annular conduit, as described in the document FR-A1-2 981 989.

(51) FIG. **10** illustrates an alternative embodiment of the assembly **62** in which the upstream segment **62a** of the O-Duct type is associated with a downstream segment **62b** of the D-Duct type.

The segment **62b** is formed of two half-shells **62b1**, **62b2** articulated with respect to each other about an axis Z substantially parallel to the axis X. The opening of the segment **62b**, by pivoting the half-shells **62b1**, **62b2** about the axis X, allows access to the interior of the segment **62b** during a maintenance or inspection operation.

(52) FIG. **11** shows exchangers **26** housed in the duct **V12**.

(53) FIG. **12** shows equipment **64** housed in the cold compartment **C2**. The center distance between the planes **P1**, **P2** and the positioning of the connecting rods **56** at the level of the members **48** allows to free up space in the compartment **C2** for the installation of the equipment **64**. In particular, such equipment **64** may comprise electronic equipment, an accessory box or gear box, etc.

(54) In yet another variant not shown, the turbine engine **10** according to the invention could be of the type shown in FIG. **6** and comprise at least one main propeller **20**, or a propeller doublet, downstream of the gas generator **12**.

(55) This is in particular the case of the alternative embodiment shown in FIG. **13** in which the upstream suspension members **48** are located at the level of the cold compartment **C2**, the thrust-absorbing rods **56** extend from the gas generator **12** to a point of attachment to the pylon **28**, and the downstream suspension members **50** are located between the suspension members **48** and the thrust-absorbing rods **56**.

(56) The variants illustrated in FIGS. **14** to **16** show that the thrust-absorbing rods **56** can be located upstream of the splitter nose **32**, between the gas generator **12** and the pylon **28** (FIG. **14**), downstream of the splitter nose **32**, between the internal casing **54** and the pylon **28** (FIG. **15**—the plane **P3** is here receded), or downstream of the splitter nose **32**, between the internal casing **54** and the pylon **28** (FIG. **16**—the plane **P3** is here receded and almost merged with the plane **P2**).

(57) The turbine engine **10** according to the invention has many advantages among which: The cold compartment **C2** is relatively large, which is advantageous for housing the equipment **64**. Another advantage is to have a cold compartment **C2** protected from fire and thermal radiation by the duct **V12** for the flow of the flow **F12**, which allows to increase the service lifetime of the equipment **64** installed in this compartment **C2** while optimizing their design without very high temperature resistance constraints. Moreover, from an aerodynamic point of view, the diameter of the duct **V12** is relatively small compared to the prior art, which means that the radial dimension of the duct is larger for the same passage cross-section compared to the current technique, which facilitates the integration of the heat exchangers. In addition, it is easier to control the mach in the duct **V12** (the mach in the duct is directly related to the passage cross-section). Indeed, with a larger diameter duct, a very small deviation in the radial dimension of the duct has a strong impact on the passage cross section.

Claims

1. A triple-flow turbine engine for an aircraft, this turbine engine having a longitudinal axis and comprising: a gas generator comprising at least one compressor, an annular combustion chamber, and at least one turbine, at least one main propeller rotated by a shaft of the gas generator, the gas generator comprising: a first annular splitter nose for separating two annular ducts for the respective flow of an internal primary flow inside the gas generator and an external secondary flow around the gas generator, a second annular splitter nose for separating two annular ducts for the respective flow of a first internal flow and of a second external flow, a first nozzle for ejecting the first internal flow, a second nozzle for ejecting the second external flow, a first annular compartment, referred to as hot compartment, extending axially between the second splitter nose and the first nozzle, and between first and second annular walls, respectively internal and external, which extend around each other, the first wall externally defining said duct for the flow of the first internal flow, and said second wall internally defining said duct for the flow of the second external

flow, a second annular compartment, referred to as cold compartment, extending axially between the first splitter nose and the second nozzle, and between third and fourth annular walls, respectively internal and external, which extend around each other, the third wall externally defining said duct for the flow of the second external flow, and said fourth wall internally defining said duct for the flow of the secondary flow, upstream suspension members of the turbine engine, which are located in a first plane perpendicular to said axis and which are connected or attached to the gas generator, downstream suspension members of the turbine engine, which are located in a second plane perpendicular to said axis and which are connected or attached to the gas generator, and thrust-absorbing rods which comprise first ends connected or attached to the gas generator and opposite second ends located in a third plane perpendicular to said axis, wherein said first, second and third planes are located axially between the first splitter nose and the second nozzle, and in that at least one part of said upstream and downstream suspension members are connected or attached to said third wall.

2. The turbine engine according to claim 1, wherein the downstream suspension members are attached or connected to said third wall.
 3. The turbine engine according to claim 1, wherein the downstream suspension members are located in the cold compartment.
 4. The turbine engine according to claim 1, wherein the upstream suspension members are attached to the third wall and/or to the fourth wall.
 5. The turbine engine according to claim 4, wherein the upstream suspension members are located just downstream of said first splitter nose.
 6. The turbine engine according to claim 1, wherein said thrust-absorbing rods extend from the upstream suspension members or downstream of these members to connection points on the gas generator, these connection points being located inside the third wall.
 7. The turbine engine according to claim 1, wherein at least one part of said second and third walls are connected together by tubular reinforcements or arms, to form an assembly.
 8. The turbine engine according to claim 7, wherein the second and third walls comprise openings and removable hatches for closing these openings.
 9. The turbine engine according to claim 7, wherein said assembly forms a segment in one piece.
 10. The turbine engine according to claim 7, wherein said assembly comprises a first segment in one piece and a second segment which can be detached from the first segment and which can itself be in one piece or formed of two half-shells articulated with respect to each other about an axis substantially parallel to said longitudinal axis.
 11. The turbine engine according to claim 1, wherein at least one heat exchanger is located in said second external flow duct, and/or at least one equipment, such as an accessory box or gearbox, is located in said cold compartment.
 12. The turbine engine according to claim 1, wherein said at least one main propeller is located upstream of the gas generator.
 13. The turbine engine according to claim 1, wherein said at least one main propeller is located downstream of the gas generator.
 14. The turbine engine according to claim 1, wherein it comprises a secondary propeller driven in rotation by a shaft of the gas generator, this propeller being located in the duct for the flow of the primary flow and upstream of said second annular splitter nose.
 15. An assembly comprising a turbine engine according to claim 1, and a pylon for hanging this turbine engine to an aircraft, the pylon having a generally elongate shape and being connected or attached respectively to the upstream and downstream suspension members of the turbine engine.
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